LaTeX for Physics
Reports, Equations, Figures, and References

UCC Physics Society

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1 One shared document-writing guide

LaTeX turns plain-text source into a typeset document. Edit `.tex`, not the generated PDF. The syntax is the same on Windows, macOS, and Linux; installation is in the OS guides. Use LaTeX for equations, consistent structure, cross-references, and bibliographies rather than arranging every page manually. For an official introduction and reference material, see the Overleaf LaTeX introduction and the LaTeX Project documentation.

2 Overleaf or local TeX?

Overleaf: Browser-based compilation and collaboration without installing a local TeX distribution. Requires network/account; check sharing settings.

Local editor and TeX: Offline builds and Git-friendly source, with separate editor and TeX installations.

For a new Overleaf project, use New Project > Blank Project and follow its project instructions. Do not upload confidential research without permission. For local work, build this guide from the repository root:

```
latexmk -cd -pdf -interaction=nonstopmode -halt-on-error LaTeX/LaTeX.tex
```

The `latexmk` command runs the necessary LaTeX and BibTeX passes.

3 Guide and example files

This guide is self-contained: copy its entire source into a new `.tex` file or an Overleaf blank project and compile it. No other workshop files are required. A local build needs a TeX distribution with the packages listed at the top of the source.

`LaTeX.tex / LaTeX.pdf:` This guide and its compiled PDF.

`ref.bib:` Optional example bibliography for practising BibTeX.

`LaTeX_meme.png:` Optional image for practising image inclusion.

Files such as `.aux`, `.log`, `.toc`, `.out`, `.fls`, and `.fdb_latexmk` are generated build state. `.bbl` is generated bibliography content; `.synctex.gz` links source and PDF positions. Do not edit these by hand.

4 Text, structure, and lists

```
\section{Method}
Normal text, \textbf{bold}, and \emph{emphasis}.

A blank line starts a new paragraph.
\begin{itemize}
  \item First point.
  \item Second point.
\end{itemize}
```

Commands start with a backslash; required arguments use braces. Environments have matching begin/end commands. A percent sign starts a source comment. Literal special characters include `\%`, `\_`, `\&`, and `\#`. Do not make paragraphs with repeated `\\` or align equations using spaces. For a local size change, group it: `{\large Larger text}`.

4.1 Formatting, alignment, and font sizes

The original tutorial's formatting examples can be written as:

```
\textbf{Bold text}, \textit{italic text}, \underline{underlined text}.
\begin{center}
  Centred text.
\end{center}
\begin{flushright}
  Right-aligned text.
\end{flushright}
\begin{flushleft}
  Left-aligned text.
\end{flushleft}
{\Huge Very large text}\par
{\LARGE Larger text}\par
{\Large Large text}\par
{\normalsize Normal text}\par
{\small Small text}\par
{\footnotesize Footnote-sized text}\par
{\scriptsize Script-sized text}\par
{\tiny Tiny text}\par
```

Braces keep a size declaration local. Prefer the document's normal styles for reports rather than manually resizing each paragraph. Use `\begin{enumerate}` with `\item` for a numbered list. For a two-column section use `\twocolumn` and later `\onecolumn`; these commands normally start a new page, so use them deliberately.

5 Equations and references

```
Inline mathematics: $F=ma$.
\begin{equation}
  T = 2\pi\sqrt{\frac{L}{g}}
  \label{eq:period}
\end{equation}
See Equation~\ref{eq:period}.
```

Numbered equations can be referenced without manually entering their numbers. Use `align` from `amsmath` for aligned multi-line work:

```
\begin{align}
  F &= ma,\\
  E &= \frac{1}{2}mv^2.
\end{align}
```

Keep units upright, for example `\(L=0.50\,\mathrm{m}\)`. For advanced units learn `siunitx`; check package compatibility before adding it to older templates. State model assumptions, not just equations.

5.1 More physics notation

These examples preserve the original tutorial's wider mathematical coverage using standard `amsmath`/`amssymb` notation; the `physics` package is not required.

```
E = mc^2
\left\langle \psi_i \mid \psi_j \right\rangle = \delta_{ij}
\mathbf{u}\cdot\mathbf{v} = \sum_{i=1}^{n}u_i v_i
\frac{dy}{dx}, \quad \frac{\partial f}{\partial x}
\int_0^1 x^2\,dx
```

```
\iiint_V f(x,y,z)\,dx\,dy\,dz
\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}
\mathbf{F} = m\mathbf{a}
\nabla\cdot\mathbf{E} = \frac{\rho}{\epsilon_0}
c = 2.99792458\times10^8\,\mathrm{m\,s^{-1}}
```

Put these expressions inside maths delimiters. For example:

$$\langle \psi_i \mid \psi_j \rangle = \delta_{ij}, \qquad \mathbf{u}\cdot\mathbf{v} = \sum_{i=1}^{n} u_i v_i, \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

$$\frac{dy}{dx}, \qquad \frac{\partial f}{\partial x}, \qquad \int_0^1 x^2\,dx, \qquad \iiint_V f(x,y,z)\,dx\,dy\,dz.$$

Use explicit bounds or a named integration domain. The small space `\,` before a differential improves readability. For structured unit formatting, the optional siunitx package provides `\SI`.

5.2 Source-code listings

Use `listings` to typeset code rather than inserting a screenshot. The following example demonstrates the source of a listing:

```
\begin{lstlisting}[language=Python]
import numpy as np
x = np.linspace(0, 10, 100)
print(x)
\end{lstlisting}
```

Follow your module's guidance about including analysis code in an appendix or separate repository. Code shown in a listing is not executed by LaTeX.

6 Figures without fighting floats

First generate `fit_result.pdf` using the Python notebook. With `\usepackage{graphicx}` in the preamble:

```
\begin{figure}[htbp]
  \centering
  \includegraphics{../Example_Code/fit_result.pdf}
  \caption{Synthetic position data and a linear model.
           Error bars show assumed position uncertainty.}
  \label{fig:fit}
\end{figure}
Figure~\ref{fig:fit} compares observations and model.
```

For Overleaf, upload the figure and change the path to `fit_result.pdf`. Prefer vector PDF for plots; use suitable raster formats for photographs. The path is relative to this guide's directory; filenames and case must match. In Overleaf, upload the figure and use its project-relative path.

A figure is a **float**. `[htbp]` suggests placements, not an exact position. Reference the figure in text; do not force every figure with `[H]`. Put the label *after* the caption. Captions should explain quantities, units, error bars, and physical meaning. See Overleaf's guide to figures and images.

6.1 An image example you can rebuild

To practise including an external image, copy the optional `LaTeX_meme.png` beside your source (or upload it to Overleaf). Then try this example:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.45\linewidth]{LaTeX_meme.png}
  \caption{An existing tutorial image used to demonstrate image inclusion.}
  \label{fig:tutorial-image}
\end{figure}
```

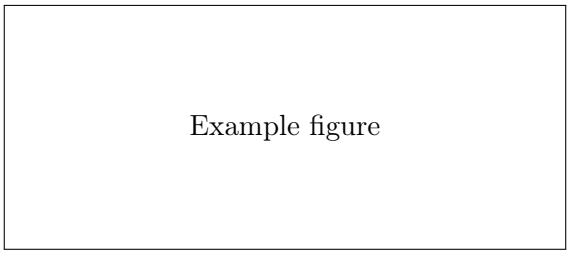


Figure 1: A self-contained placeholder demonstrating figure sizing and references.

Figure 1 uses a relative width so it scales with the text area. The `[H]` placement option in the old tutorial requires `float`; it prevents floating, so use it only when genuinely necessary.

7 Tables and citations

Use tables for numerical results, with units in headings. `booktabs` provides `\toprule`, `\midrule`, and `\bottomrule`. Round results to sensible significant figures.

```
\begin{table}[htbp]
  \centering
  \caption{Illustrative values for practising table layout.}
  \label{tab:example}
  \begin{tabular}{rr}
    \toprule
    $x$ & $y$ \\
    \midrule
    1 & 2 \\
    2 & 4 \\
    \bottomrule
  \end{tabular}
\end{table}
```

For a separate BibTeX bibliography, create `ref.bib` beside your source with this entry:

```
@misc{overleaf_examples,
  author = {{Overleaf}},
  title = {Overleaf LaTeX Examples},
  year = {2024},
  howpublished = {\url{https://www.overleaf.com/latex/examples}}
}
```

In your document, cite it and print the bibliography with:

```
See the examples~\cite{overleaf_examples}.
\bibliographystyle{plain}
\bibliography{ref}
```

The citation key must match the BibTeX entry exactly. See Overleaf's BibTeX guide for citation and bibliography syntax. Check source metadata and whether the source supports your claim. A formatted citation is not proof of correctness. For a complete rendered citation example in this guide, see the Overleaf examples collection [1]. This guide instead uses an inline

`thebibliography` environment, so its source compiles without a separate `.bib` file or BibTeX. Do not combine the two bibliography approaches in the same document. For a clickable external link, use `\href{URL}{link text}`.

8 A useful lab-report structure

1. Aim and physical model, including assumptions.
2. Method, instrument calibration, and data collection.
3. Results with units and clear figures/tables.
4. Uncertainties, residuals, systematic effects, and limitations.
5. Conclusion supported by measurements.
6. References and appropriate supplementary analysis/code.

Follow your module's rubric when it differs.

9 Exercises and checks

1. Change the title/author and rebuild; confirm the PDF changed.
2. Add a labelled equation and refer to it without typing its number.
3. Add a table row, clearly marking real versus synthetic data.
4. Insert the Python plot and write a meaningful caption.
5. Add a real source, cite it, and rebuild without question marks.
6. Reorder two figures and check that references still follow them.

10 Troubleshooting

No PDF: Read the first error, not just the last summary.

Undefined control sequence: Check spelling and required packages.

Missing dollar inserted: Check maths delimiters and ordinary-text underscores.

Figure missing: Check working directory, filename/case, and whether the figure was generated.

Citation question mark: Check keys, bibliography name, BibTeX, and two further LaTeX passes when using BibTeX. For this guide's inline bibliography, rerun LaTeX to resolve citations.

Figure moved: This is normal float behaviour; use cross-references.

Overfull box: Inspect the PDF; shorten text/URLs or restructure maths.

Missing package: Use the TeX distribution manager or Overleaf; an editor extension is not TeX itself.

Further reference: Overleaf learning resources and LaTeX documentation.

References

[1] Overleaf. *Overleaf LaTeX Examples*. 2024. https://www.overleaf.com/latex/examples.